Solder Paste Stencil Design

Apertures • Thickness • Step stencils • BGA reductions

The thin steel sheet that defines every solder joint

Audience: Engineers preparing for SMT assembly

What • How made • Aperture sizing • Fine pitch • BGA • Defects

What is a Stencil

- Thin steel (or nickel) sheet with cut-out apertures matching every pad on the PCB
- Placed over bare PCB; solder paste forced through openings by squeegee
- When stencil lifted, paste deposits remain on every pad
- Components then placed on the paste; reflow melts paste → joints
- Typical thickness: 0.1-0.15 mm (4-6 mil)
- Aperture sizes: typically same as pad (1:1) or slightly smaller for fine pitch
- One stencil per PCB design (and rev) — cost ₹500-1500 per stencil

Stencil Manufacturing Methods

Laser-cut

Most common today.
Sharp clean walls.
Fast turnaround.

Chemical etch

Older method.
Tapered walls (release benefit).
Cheaper at low volume.

Electroformed

Premium — for ultra-fine pitch.
Ultra-smooth walls.
5-10× cost.

Step stencil

Different thicknesses
in different regions.
Used for mixed pitch.

Framed

Stencil mounted in metal frame.
Standard for production.

Frameless

Loose foil for prototype.
Cheap, less precise.

Aperture Sizing Rules

- Standard rule: aperture = pad size (1:1 ratio)
- Fine pitch (< 0.5 mm): reduce aperture 10-20% to avoid bridging
- BGA: aperture = ball pad diameter × 0.9 (smaller stencil aperture)
- Aspect ratio: aperture width / stencil thickness should be > 1.5
- Area ratio: (aperture area) / (aperture wall area) should be > 0.66
- Aspect/area ratios determine paste release quality
- For thinner stencils (0.1 mm), small apertures release better than thicker

Stencil Thickness Selection

Smallest pitch	Recommended thickness
≥ 0.8 mm	0.15 mm (6 mil)
0.65 mm	0.125 mm (5 mil)
0.5 mm	0.1 mm (4 mil)
0.4 mm	0.075-0.1 mm (3-4 mil)
0.3 mm BGA	0.075 mm (3 mil)
0.2 mm fine pitch	0.05-0.075 mm (2-3 mil)

Mixed-pitch boards

If you have both 0.8 mm and 0.4 mm pitch components on one board, use a STEP STENCIL — thicker in one area, thinner in another. Cost premium ~30-50%.

Aperture Shape — Square vs Rounded

- Default: aperture shape matches pad shape (rectangle for rectangular pads, circle for round)
- Rectangle apertures: fine-pitch ICs (TQFP, QFN)
- Trim corners (chamfered apertures) on rectangle to reduce bridging
- Round apertures: BGA balls, vias-in-pad
- 'D-shaped' apertures: combine paste from adjacent pads (use sparingly)
- Home-plate (pentagon) aperture: directs paste flow on QFN — minor benefit

BGA Stencil Considerations

- BGA aperture = ball pad × 0.85-0.9 (smaller stencil opening)
- Reason: BGA pads are smaller than passive pads; risk of bridging is high
- Standard 0.4 mm pitch BGA: aperture ~0.21 mm dia, stencil 0.075 mm thick
- 0.3 mm pitch (FBGA mobile parts): require 0.05 mm stencil + electroformed for paste release
- Step stencil region around BGA: 0.075 mm in BGA, 0.125 mm in surrounding area
- Avoid mixing BGAs of dramatically different pitches without step stencil

Paste Volume Control

- Paste volume = aperture area × stencil thickness
- Too much: bridges, solder balls, tombstoning
- Too little: open joints, weak fillets
- SPI (Solder Paste Inspection) measures actual deposited volume
- Target: 90-110% of nominal aperture × thickness
- Reduce volume on small SMD passives (0402, 0201): 60-70% of pad area
- Increase volume on thermal pads: full pad area or use 'window' pattern

Thermal Pad Aperture Design

- QFN / DFN with center thermal pad: don't paste 100% of pad
- Too much paste under center pad → IC floats during reflow → joint defects
- Recommended: 50-70% paste coverage on thermal pad
- Pattern options:
- • Grid of small apertures (4×4 or 6×6 squares)
- • 'Window-pane' grid (single large opening with cross dividers)
- • Single smaller aperture (70-80% of pad)
- Vendor datasheets specify recommended thermal pad stencil design

Common Paste Defects + Fixes

Defect	Cause	Stencil-side fix
Solder bridges	Aperture too large	Reduce aperture 10-15%
Insufficient solder	Aperture too small / clogged	Increase aperture / clean
Tombstoning	Asymmetric pad paste	Symmetric apertures, equal volume
Solder beading	Paste squeezed out	Reduce aperture or step stencil
BGA voids	Paste outgassing	Aperture pattern with vent gaps
Open joints	No paste deposited	Check aperture size, stencil flatness
Skipped paste	Clogged stencil aperture	Clean stencil mid-run

Stencil Procurement

- Buy with the PCB order: JLCPCB / PCBWay offer stencils as add-on (~₹500)
- Standalone stencils: OSH Stencils ($25), Pololu ($30+), Indian fabs
- File format: provide Gerber 'cream mask' layer (the paste layer from your EDA tool)
- Specify: thickness (default 0.12 mm), framed or frameless, size
- Lead time: 1-3 days (same as PCB for combined orders)
- Storage: keep flat, dust-free, refrigerate if combined with paste
- Lifespan: 50-200 print cycles with care; longer with electroformed

DFM Tips for Stencil

- Don't put fine-pitch (BGA) next to standard-pitch (0805) on same side — stencil compromise
- Use vendor-recommended stencil design for QFN thermal pads
- Verify aperture sizes via fab DFM check before ordering
- Add fiducials to stencil too (matching panel fiducials) for printer alignment
- Specify stencil rev when you respin a PCB
- Test stencil with PASTE before placing IC — visually check paste deposits
- For low-volume: hand-stencil OK with a squeegee + practice

Key Takeaways

Key takeaways from this lecture

Aperture = pad

Reduce for fine pitch.

Thickness by pitch

Smaller pitch = thinner stencil.

BGA reduction 10-15%

Smaller aperture vs ball pad.

Thermal pad ≤ 70%

Avoid IC floating in reflow.

Step stencil for mixed

Thicker + thinner regions.

SPI catches defects

Measure paste before P&P.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- IPC-7525 — Stencil Design Guidelines
- JLCPCB / PCBWay stencil add-on offers
- OSH Stencils + Pololu — independent stencil services
- Indium / Kester paste vendor app notes on stencil design
- BGA assembly guides from Texas Instruments / Microchip
- Phil's Lab — stencil + reflow tutorials
- Indian stencil sources — local PCB fabs
- Step-stencil case studies (Heraeus, AIM)

Thank you • Build something this week.